IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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Date: 6 Mar 2026

Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
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Executive Summary

Summary of Methodologies

1. Data Collection: Accessed SpaceX launch data via API and web scraped records from Wikipedia.

2. Data Cleaning & Preparation:
- Cleaned and formatted the data.
- Stored data in Db2 database and performed SQL queries.
- Conducted exploratory data analysis.

3. Feature Engineering: Created new features and standardized the data.

4. Interactive Visualizations:
- Mapped launch sites and success rates using Folium.
- Built an interactive dashboard with Plotly Dash.

Executive Summary

5. Model Building & Evaluation:
- Implemented SVM, Decision Trees, and K-Nearest Neighbors.
- Tuned hyperparameters with GridSearchCV.
- Evaluated models using test data accuracy.

Summary of Results

1. Data Insights:
- Identified factors influencing Falcon 9 first stage landings.
- Visualized geographical patterns and success rates.

2. Model Performance:
- SVM and K-Nearest Neighbors: 83.33% accuracy.
- Decision Tree: 94.44% accuracy.

3. Key Findings:
- Launch site and payload mass impact landing success.
- Decision Tree model is the most effective predictor.

Introduction

Project background and context

The primary objective of this capstone project is to forecast the successful landing of the Falcon 9's first stage. SpaceX maintains a competitive edge by offering launch services at a fraction of the cost of its rivals, a feat made possible through the innovative reuse of their first-stage boosters. By developing a model that accurately predicts landing outcomes, we can effectively estimate launch expenses and deliver strategic intelligence for organizations competing for launch contracts against SpaceX.

Problems you want to find answers

- What factors influence the successful landing of the Falcon 9 first stage?
- How can we accurately predict the landing outcome using machine learning models?
- Which machine learning model performs best in predicting the landing success?

Section 1

Methodology

Methodology

Executive Summary: This project employs a comprehensive approach to predict the successful landing of the Falcon 9 first stage, incorporating data collection, processing, exploratory analysis, interactive visualizations, and predictive modeling.

- **Data collection methodology**: Data was sourced from the SpaceX API, which provided detailed records of Falcon 9 launches, including launch dates, sites, payloads, and outcomes.
- **Perform data wrangling**: Data cleaning involved handling missing values, standardizing formats, and ensuring consistency. Key features were extracted and new features engineered to enrich the dataset.
- **Perform exploratory data analysis (EDA) using visualization and SQL**:
  - Visualized launch success rates, payloads, and launch sites using Matplotlib and Seaborn.
  - Executed SQL queries to derive insights and answer specific questions regarding the dataset

Methodology

- **Perform interactive visual analytics using Folium and Plotly Dash:**
  - Used Folium to create interactive maps displaying launch sites and outcomes.
  - Developed a Plotly Dash application with interactive components like dropdowns and sliders to analyze launch success rates and payload ranges.
- **Perform predictive analysis using classification models:**
  - Built and evaluated various classification models including Logistic Regression, SVM, KNN, and Decision Trees.
  - Employed GridSearchCV for hyperparameter tuning.
  - Evaluated models based on accuracy, and identified the best performing model for predicting landing success.
- Github URL: https://github.com/datlt-edulive/Data-Science-Capstone-Project

Data Collection

- Step 1: SpaceX API request

- Step 2: Web Scraping Wikipedia

- Step 3: Data Integration

Data Collection – SpaceX API

Step 1: Initiate API Request

- Use Python's `requests` library to connect to the SpaceX API.
- Endpoint: `https://api.spacexdata.com/v4/launches`

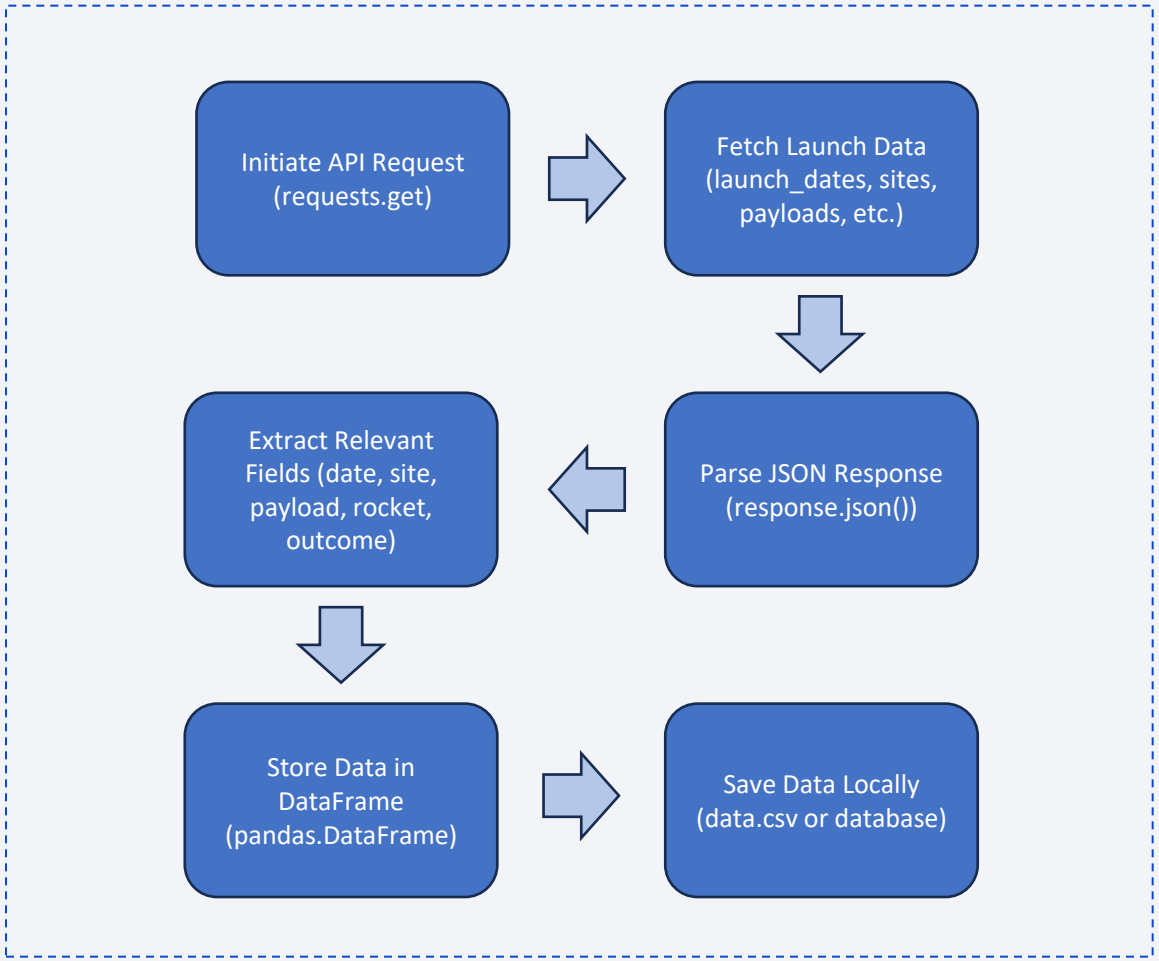
Step 2: Parse API Response

- Convert API response from JSON to a Python dictionary.
- Extract relevant fields: launch date, launch site, payload mass, rocket type, outcome.

Step 3: Store Data Locally

- Save extracted data into a pandas DataFrame.
- Store the DataFrame locally for further processing.

GitHub URL: https://github.com/datIt-edulive/Data-Science-Capstone-Project/blob/main/1.jupyter-labs-spacex-data-collection-api.ipynb



Data Collection - Scraping

Step 1: Initiate Web Scraping

- Use Python's `requests` library to fetch the HTML content of the Wikipedia page.
- Target URL: `https://en.wikipedia.org/wiki/List_of_Falcon_9_and_Falcon_Heavy_launches`

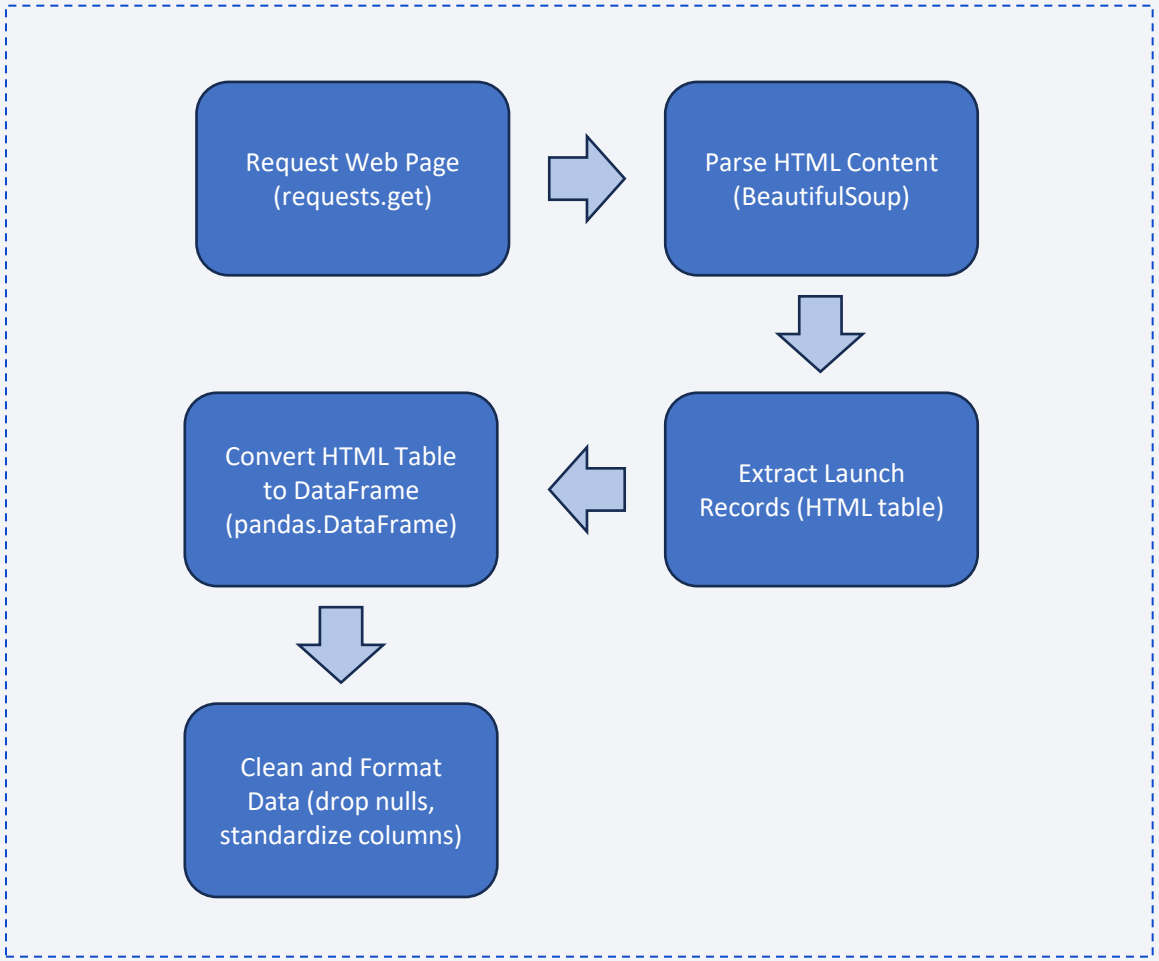
Step 2: Parse HTML Content

- Use `BeautifulSoup` to parse the HTML content.
- Extract the HTML table containing Falcon 9
- launch records..

Step 3: Convert to DataFrame

- Convert the extracted HTML table into a pandas DataFrame.
- Clean and format the DataFrame, ensuring data consistency.

GitHub URL: https://github.com/datIt-edulive/Data-Science-Capstone-Project/blob/main/2.jupyter-labs-webscraping.ipynb



Data Wrangling

Overview: Data wrangling involves cleaning, transforming, and organizing raw data into a structured format suitable for analysis

Step 1: Data Cleaning

- Identify and fill or remove missing values in the dataset.
- Use appropriate imputation techniques or drop rows/columns with excessive missing data.

Step 2: Data Transformation

- Convert data types to appropriate formats (e.g., date-time, numerical).
- Standardize text (e.g., lowercase, remove whitespace).
- Create new features from existing data (e.g., extract year from date).
- Normalize/scale numerical features to ensure consistency.

Data Wrangling

Step 3: Data Integration

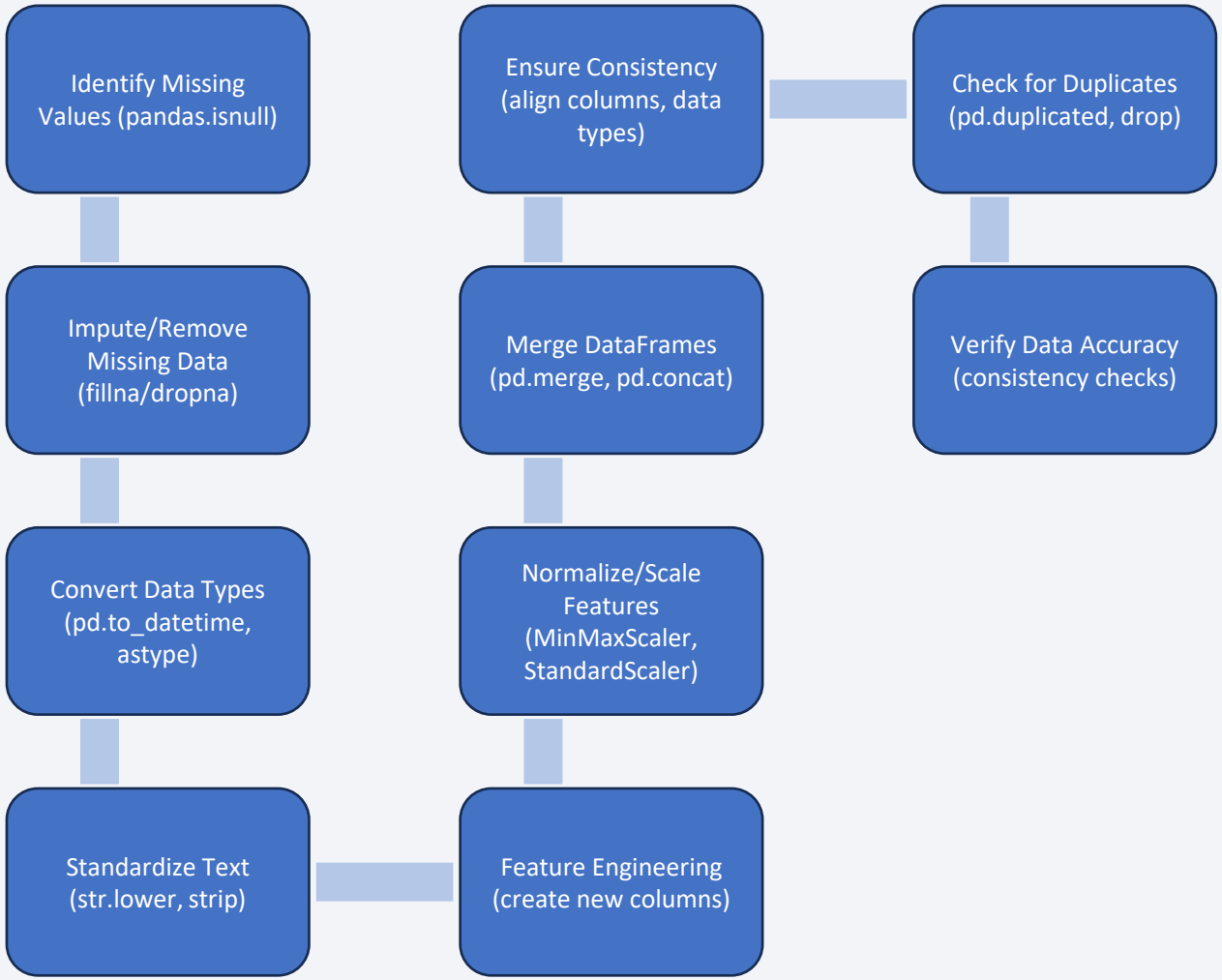
- Merge datasets collected from different sources (API, web scraping) into a single cohesive dataset.
- Ensure consistent column names and data formats across datasets.

Step 4: Data Validation

- Check for duplicate records and remove them.
- Verify the accuracy and consistency of data entries..

- Github URL: https://github.com/datlt-edulive/Data-Science-Capstone-Project/blob/main/3.labs-jupyter-spacex-Data%20wrangling.ipynb

Data Wrangling (Flowchart)



EDA with Data Visualization

Overview: Exploratory Data Analysis (EDA) involves visually exploring and summarizing the main characteristics of a dataset. The goal is to understand the data's distribution, identify patterns, and uncover relationships between variables.

Charts Plotted:

1. Histograms:

- **Purpose:** Used to visualize the distribution of numerical variables such as launch success rates, payload mass, and flight number.
- **Why:** Helps in understanding the spread and central tendency of the data, identifying outliers, and assessing data skewness.

2. Bar Charts:

- **Purpose:** Used to compare categorical variables such as launch outcomes (success/failure) across different categories like launch sites or rocket types.
- **Why:** Provides a clear comparison of frequencies or proportions within categorical data, highlighting patterns or trends.

3. Line Charts:

- **Purpose:** Used to track trends over time, such as the success rate of Falcon 9 launches across different years.
- **Why:** Reveals temporal patterns and helps in understanding performance trends or changes over specific periods.

EDA with Data Visualization

4. Scatter Plots:
- **Purpose:** Used to explore relationships between two numerical variables, such as payload mass vs. launch success.
- **Why:** Identifies correlations or dependencies between variables, visualizing how one variable changes concerning another.

5. Heatmaps:
- **Purpose:** Used to visualize correlation matrices between multiple numerical variables.
- **Why:** Helps in identifying strong correlations (positive or negative) between variables, aiding feature selection or understanding multicollinearity.

6. Box Plots:
- **Purpose:** Used to display the distribution of numerical data through their quartiles.
- **Why:** Visualizes the spread and skewness of data, highlighting outliers and comparing distributions across different categories.

Github URL: https://github.com/datlt-edulive/Data-Science-Capstone-Project/blob/main/5.%20jupyter-labs-eda-dataviz.ipynb

EDA with SQL

Aggregate Queries:
- Calculated total number of launches.
- Counted successful and failed launches.
- Calculated success rates by launch site and rocket type.

Join Queries:
- Joined tables to link launch records with additional data (e.g., rocket details).
- Combined datasets for comprehensive analysis.

Filtering Queries:
- Filtered data to focus on specific launch outcomes (success/failure).
- Applied conditions to extract launches based on criteria like launch date or rocket configuration.

Sorting Queries:
- Sorted data to identify trends or outliers.
- Ordered launches by date or success rate for analysis.

Subqueries:
- Nested queries to calculate derived metrics (e.g., average payload mass per launch site).
- Subqueries used to perform detailed analysis within larger datasets.

Github URL: https://github.com/datlt-edulive/Data-Science-Capstone-Project/blob/main/4.jupyter-labs-eda-sql-coursera_sqllite.ipynb

Build an Interactive Map with Folium

Map Objects Created

Markers:
- Placed markers to indicate launch sites on the map.
- Each marker represents a specific geographical location where SpaceX launches have occurred.

Circles:
- Added circles around launch sites to visually represent proximity zones.
- Circles help visualize the areas around launch sites that might influence operational decisions.

Lines:
- Drew lines to connect launch sites with their proximities or other relevant locations.
- Lines provide spatial context and connections between different points of interest related to launches.

Github URL: https://github.com/datlt-edulive/Data-Science-Capstone-Project/blob/main/6.%20lab_jupyter_launch_site_location.ipynb

Reasons for Adding Objects

Markers:
- To pinpoint exact launch locations for spatial reference.
- Helps users identify where SpaceX has conducted launches geographically.

Circles:
- Illustrates the potential impact zones around launch sites.
- Provides a visual representation of safety perimeters or operational boundaries.

Lines:
- Shows connections or relationships between launch sites and relevant features.
- Enhances understanding of spatial relationships and dependencies.

Build a Dashboard with Plotly Dash

Plots/Graphs Added

Success Pie Chart:

- Displays the distribution of successful and failed launches.
- Helps visualize the overall success rate and performance trends.

Success-Payload Scatter Plot:

- Shows the relationship between payload mass and launch success.
- Allows users to explore how payload mass influences mission outcomes.

Github URL: https://github.com/datlt-edulive/Data-Science-Capstone-Project/blob/main/spacex_dash_app.py

Interactions Added

Launch Site Dropdown:

- Enables users to select specific launch sites for analysis.
- Facilitates filtering and focused exploration based on geographical locations.

Range Slider for Payload:

- Allows users to adjust payload mass ranges dynamically.
- Offers flexibility in examining launch success concerning payload mass variations.

Predictive Analysis (Classification)

1. Data Processing:
- Standardized features to ensure all variables contribute equally.
- Split data into training and test sets for model validation.

2. Model Selection:
- Explored multiple classification algorithms: SVM, Decision Trees, and K-Nearest Neighbors (KNN).
- Chose algorithms suitable for binary classification tasks based on project requirements.

3. Hyperparameter Tuning:
- Used GridSearchCV to systematically search for optimal hyperparameters.
- Tuned parameters such as C (SVM), max_depth (Decision Trees), and n_neighbors (KNN).

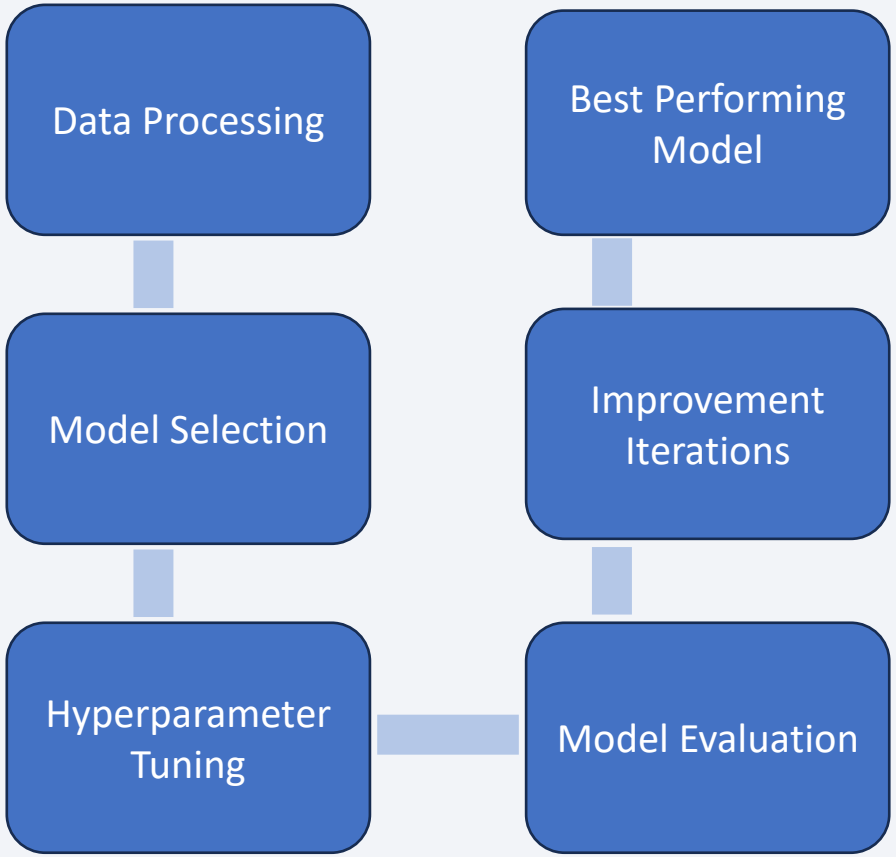
Github URL: https://github.com/datlt-edulive/Data-Science-Capstone-Project/blob/main/7.%20SpaceX_Machine_Learning_Prediction_Part_5.jupyterlite.ipynb

4. Model Evaluation:
- Evaluated models using cross-validation techniques to ensure robustness and generalizability.
- Utilized metrics like accuracy, precision, recall, and F1-score to assess model performance.

5. Improvement Iterations:
- Iteratively adjusted models based on insights from validation results.
- Fine-tuned hyperparameters to maximize predictive accuracy and reliability.

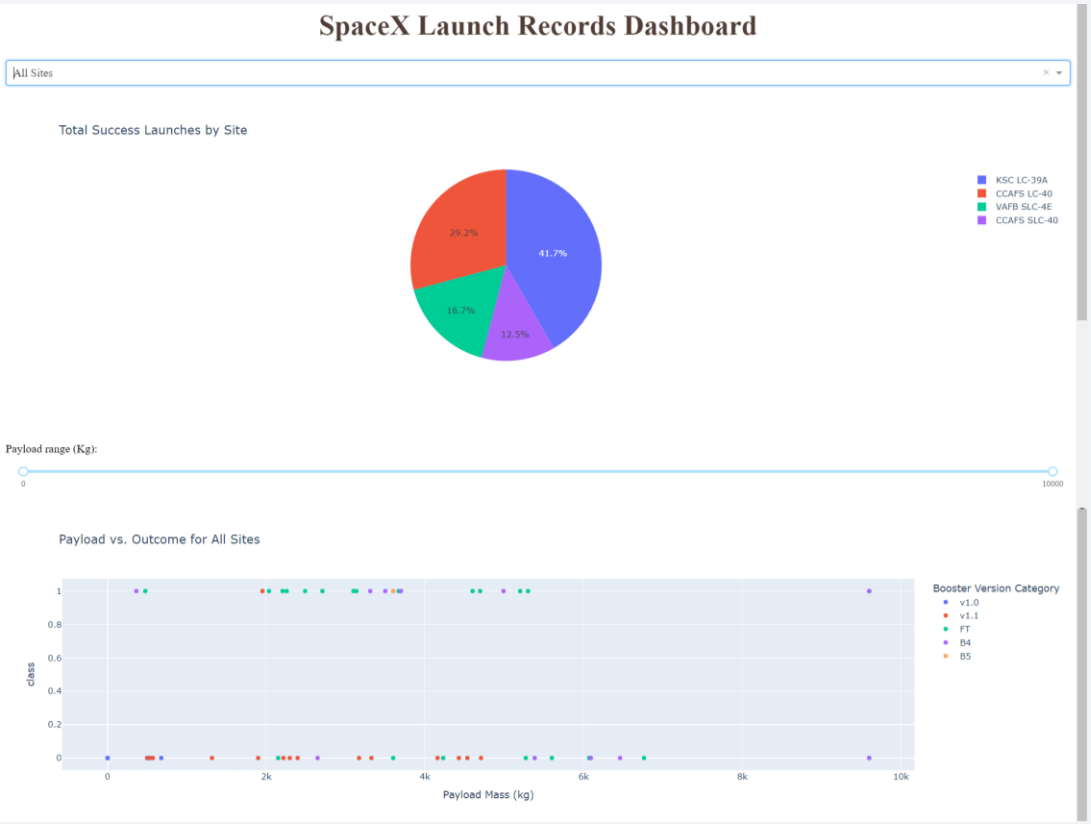
6. Selection of Best Performing Model:
- Identified the model with the highest accuracy on the test set as the best performer.
- Considered both training and test set performance to avoid overfitting and ensure real-world applicability

Predictive Analysis (Flowchart)



Results

- Interactive analytics demo in screenshots



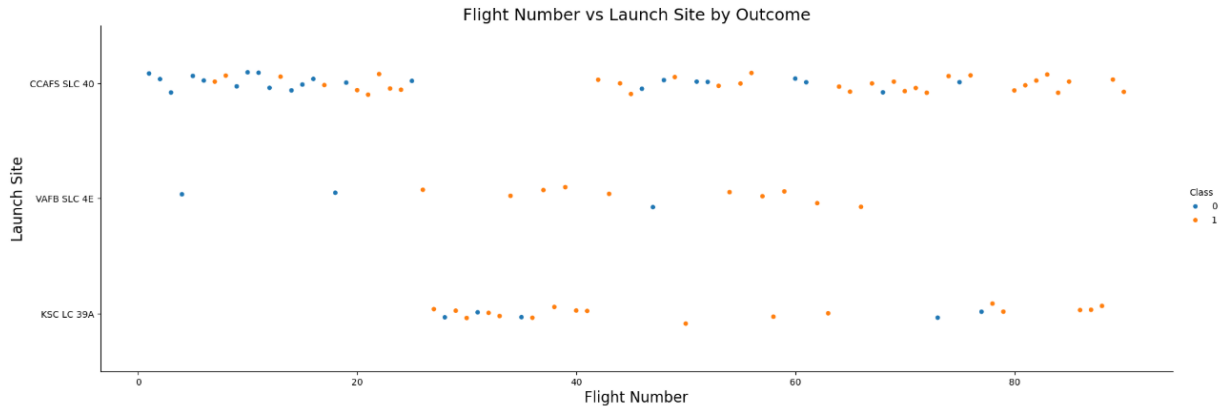
- Exploratory data analysis results

TASK 1: Visualize the relationship between Flight Number and Launch Site

Use the function `catplot` to plot `FlightNumber` vs `LaunchSite`, set the parameter `x` parameter to `FlightNumber`,set the `y` to `Launch Site` and set the parameter `hue` to `'class'`

In [10]:

```
# Plot a scatter point chart with x axis to be Flight Number and y axis to be the launch site, and hue to be the class value
sns.catplot(x="FlightNumber", y="LaunchSite", hue="Class", data=df, aspect=3, height=6, kind="strip")
plt.xlabel("Flight Number", fontsize=16)
plt.ylabel("Launch Site", fontsize=16)
plt.title("Flight Number vs Launch Site by Outcome", fontsize=18)
plt.show()
```

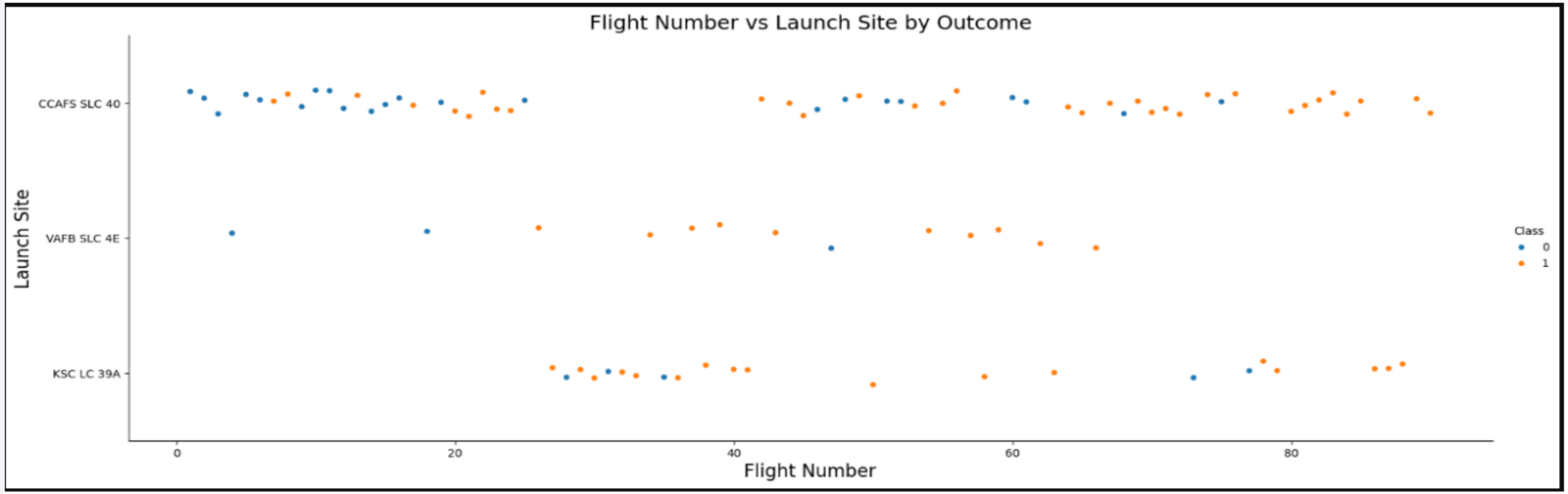


Section 2

Insights drawn from EDA

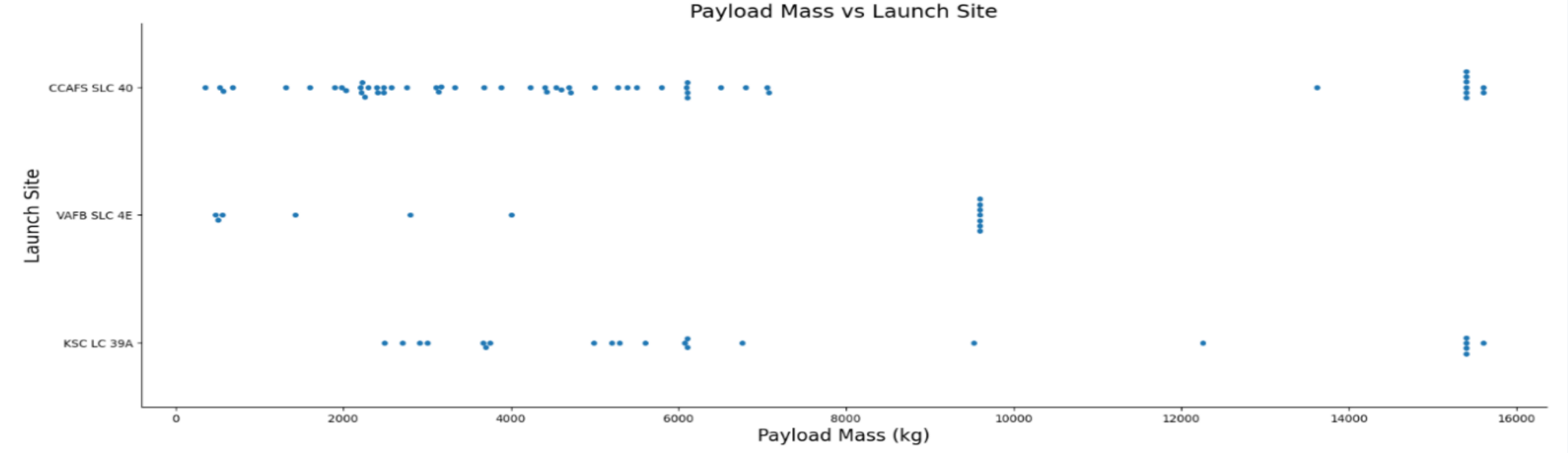
Flight Number vs. Launch Site

- **Mixed Outcomes at Major Launch Sites**: Both CCAFS SLC 40 and KSC LC 39A have a mix of successful (orange) and unsuccessful (blue) landings, indicating that factors other than the launch site itself may influence the landing success.
- **Consistent Activity Across Flight Numbers**: Launches are spread across a wide range of flight numbers at all sites, suggesting consistent activity over time without a clear trend of increasing or decreasing landing success.



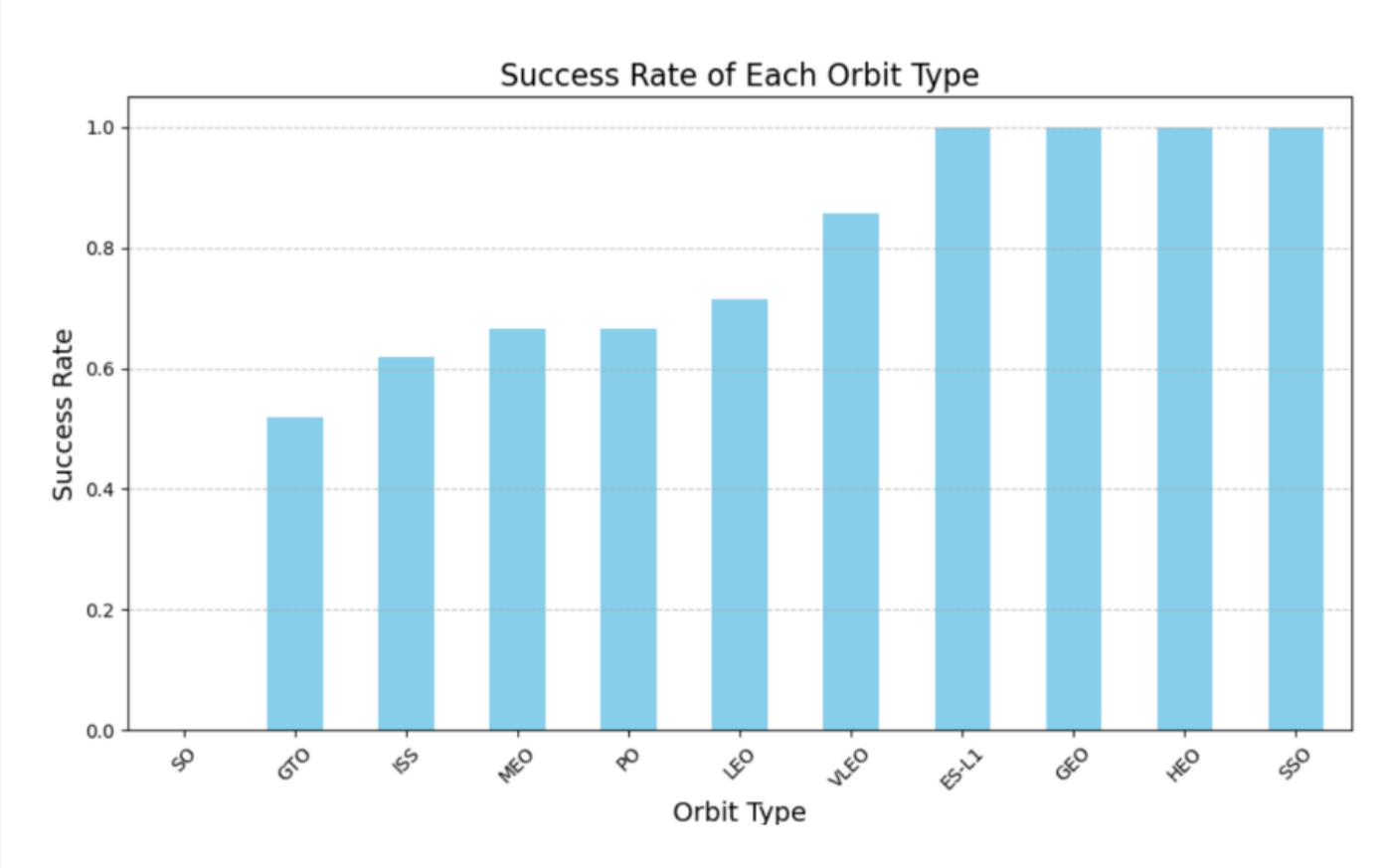
Payload vs. Launch Site

- **Payload Distribution**: Most launches from the CCAFS SLC 40 site handle payloads below 10,000 kg, while the VAFB SLC 4E and KSC LC 39A sites have a wider range of payload masses, indicating varied mission profiles.
- **High-Capacity Launches**: The KSC LC 39A site is frequently used for launching heavier payloads, with multiple launches carrying over 15,000 kg, suggesting its suitability for high-capacity missions.



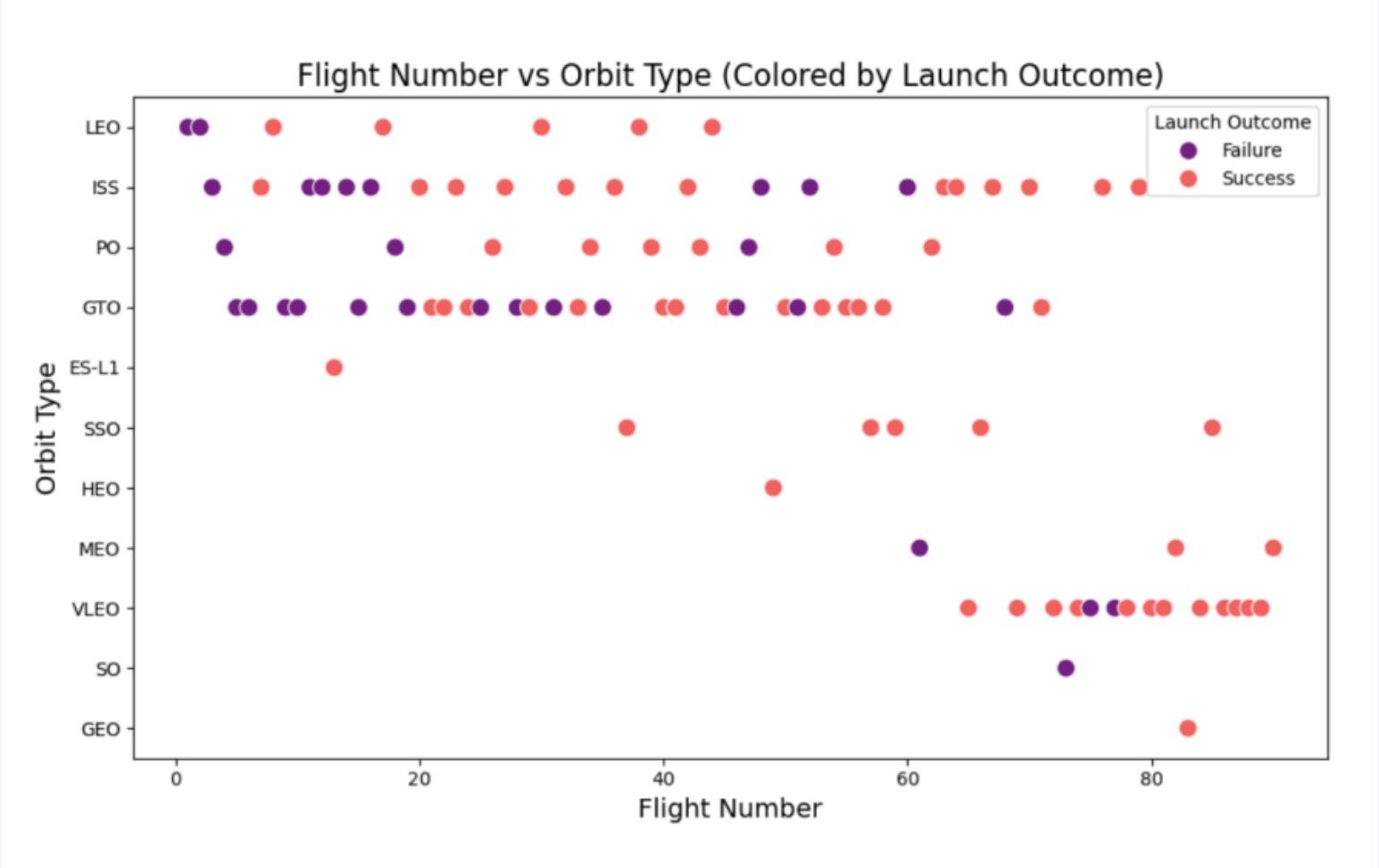
Success Rate vs. Orbit Type

- **High Success Rates**: Missions to VLEO, ES -L1, GEO, HEO, and SSO orbits have achieved a perfect success rate, indicating these orbits are highly reliable for successful first stage landings.
- **Lower Success Rate for GTO**: The GTO orbit type shows a significantly lower success rate compared to other orbit types, suggesting that missions to this orbit may involve greater challenges or complexities.



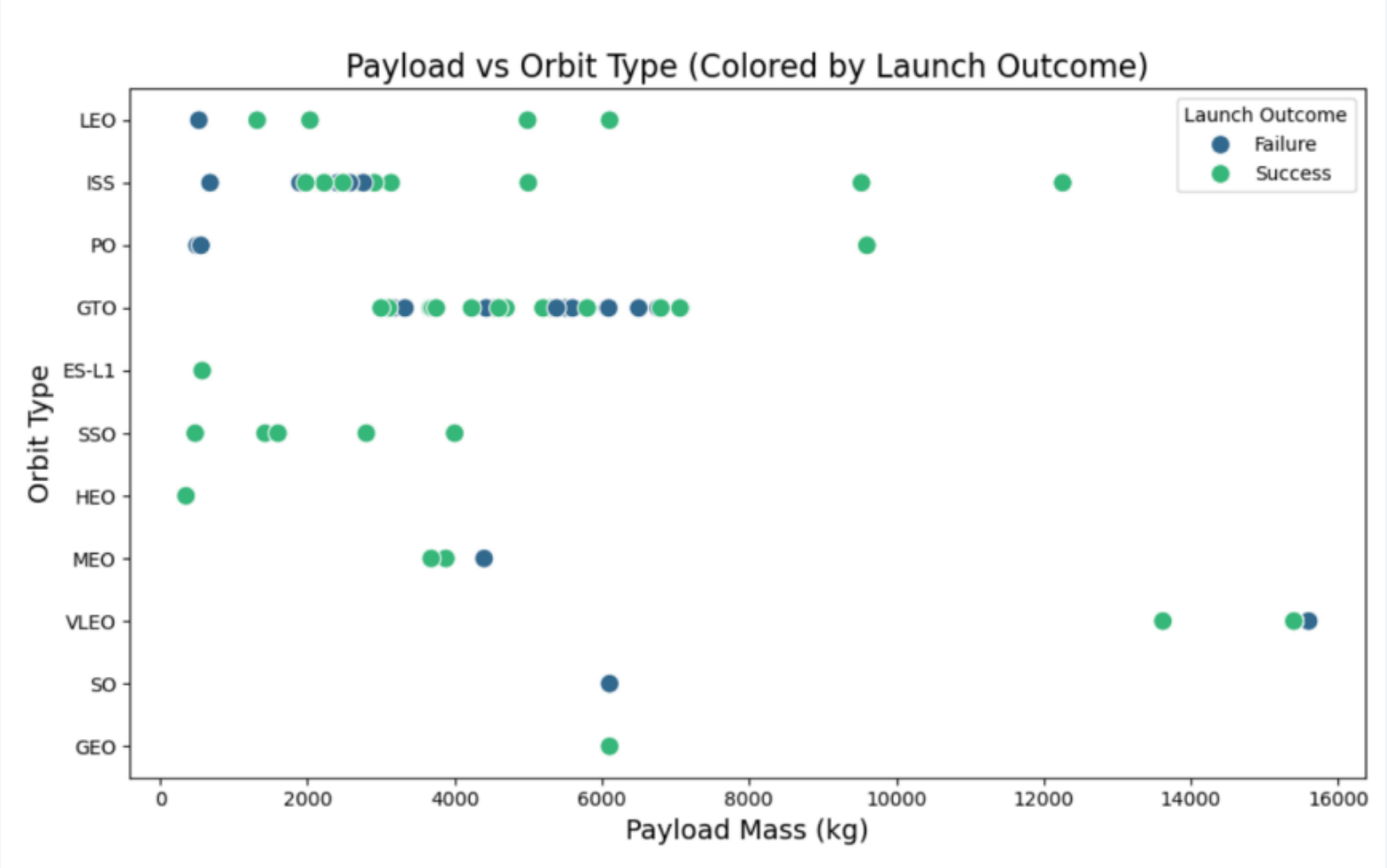
Flight Number vs. Orbit Type

- **Increased Success Over Time**: The success rate of Falcon 9 launches improves significantly with higher flight numbers, indicating that experience and iterative improvements contribute to better outcomes.
- **Orbit-Specific Performance**: Early flights to GTO and ISS orbits had mixed outcomes, but recent missions to these orbits show a higher success rate, reflecting advancements in mission planning and execution.



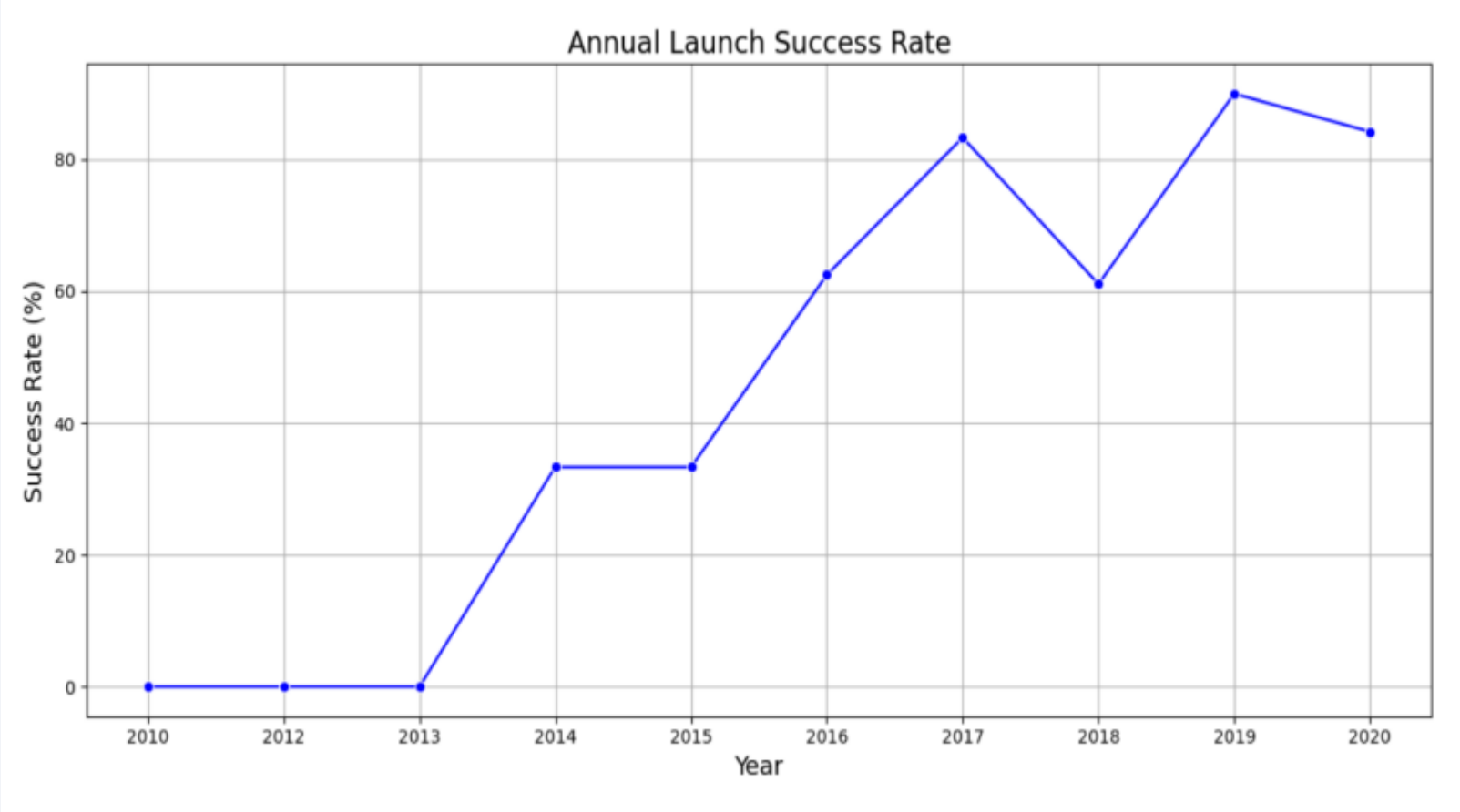
Payload vs. Orbit Type

- Successful landings are more frequent across all orbit types, especially for payloads less than 6000 kg.
- Higher payload masses (above 10,000 kg) show a mix of successes and failures, indicating increased difficulty with heavier payloads.



Launch Success Yearly Trend

- The annual launch success rate has shown a significant improvement from 2013 onwards, reaching over 80% by 2020.
- Despite a dip in 2018, the overall trend indicates increasing reliability and success in Falcon 9 launches over the years.



All Launch Site Names

Task 1

Display the names of the unique launch sites in the space mission

In [10]:
```
%sql SELECT DISTINCT "Launch_Site" FROM SPACEXTABLE;
```

```
 * sqlite:///my_data1.db
Done.
```

Out[10]:

Launch_Site
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40

Launch Site Names Begin with 'CCA'

Task 2

Display 5 records where launch sites begin with the string 'CCA'

In [11]:
```
%sql SELECT * FROM SPACEXTABLE WHERE "Launch_Site" LIKE 'CCA%' LIMIT 5;
```

```
 * sqlite:///my_data1.db
Done.
```

Out[11]:

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS__KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

Total Payload Mass

Task 3

Display the total payload mass carried by boosters launched by NASA (CRS)

In [12]:
```
%sql SELECT SUM("PAYLOAD_MASS__KG_") FROM SPACEXTABLE WHERE "Customer" = 'NASA (CRS)';
```

```
 * sqlite:///my_data1.db
Done.
```

Out[12]:

SUM("PAYLOAD_MASS__KG_")
45596

Average Payload Mass by F9 v1.1

Task 4

Display average payload mass carried by booster version F9 v1.1

In [13]:
```
%sql SELECT AVG("PAYLOAD_MASS__KG_") FROM SPACEXTABLE WHERE "Booster_Version" = 'F9 v1.1';
```

```
 * sqlite:///my_data1.db
Done.
```

Out[13]:

AVG("PAYLOAD_MASS__KG_")
2928.4

First Successful Ground Landing Date

Task 5

List the date when the first succesful landing outcome in ground pad was acheived.

Hint:Use min function

In [14]:
```
%sql SELECT MIN("Date") FROM SPACEXTABLE WHERE "Landing_Outcome" = 'Success (ground pad)';
```

```
 * sqlite:///my_data1.db
Done.
```

Out[14]:

MIN("Date")
2015-12-22

Successful Drone Ship Landing with Payload between 4000 and 6000

Task 6

List the names of the boosters which have success in drone ship and have payload mass greater than 4000 but less than 6000

In [15]:
```
%sql SELECT DISTINCT "Booster_Version" FROM SPACEXTABLE WHERE "Landing_Outcome" = 'Success (drone ship)' AND "PAYLOAD_MASS__KG_" > 4000 AND "P
```

```
 * sqlite:///my_data1.db
Done.
```

Out[15]:

Booster_Version
F9 FT B1022
F9 FT B1026
F9 FT B1021.2
F9 FT B1031.2

Total Number of Successful and Failure Mission Outcomes

Task 7

List the total number of successful and failure mission outcomes

In [16]:

```
%sql SELECT "Mission_Outcome", COUNT(*) AS "Total" FROM SPACEXTABLE WHERE "Mission_Outcome" IN ('Success', 'Failure') GROUP BY "Mission_Outcom
```

```
 * sqlite:///my_data1.db
Done.
```

Out[16]:

Mission_Outcome	Total
Success	98

Boosters Carried Maximum Payload

Task 8

List all the booster_versions that have carried the maximum payload mass, using a subquery with a suitable aggregate function.

In [17]:
```
%sql SELECT DISTINCT "Booster_Version" FROM SPACEXTABLE WHERE "PAYLOAD_MASS__KG_" = (SELECT MAX("PAYLOAD_MASS__KG_") FROM SPACEXTABLE);
```

```
 * sqlite:///my_data1.db
Done.
```

Out[17]:

Booster_Version
F9 B5 B1048.4
F9 B5 B1049.4
F9 B5 B1051.3
F9 B5 B1056.4
F9 B5 B1048.5
F9 B5 B1051.4
F9 B5 B1049.5
F9 B5 B1060.2
F9 B5 B1058.3
F9 B5 B1051.6
F9 B5 B1060.3
F9 B5 B1049.7

2015 Launch Records

Task 9

List the records which will display the month names, failure landing_outcomes in drone ship ,booster versions, launch_site for the months in year 2015.

Note: SQLLite does not support monthnames. So you need to use substr(Date, 6,2) as month to get the months and substr(Date,0,5)='2015' for year.

In [18]:

```
%%sql

SELECT
    CASE
        WHEN substr("Date", 6, 2) = '01' THEN 'January'
        WHEN substr("Date", 6, 2) = '02' THEN 'February'
        WHEN substr("Date", 6, 2) = '03' THEN 'March'
        WHEN substr("Date", 6, 2) = '04' THEN 'April'
        WHEN substr("Date", 6, 2) = '05' THEN 'May'
        WHEN substr("Date", 6, 2) = '06' THEN 'June'
        WHEN substr("Date", 6, 2) = '07' THEN 'July'
        WHEN substr("Date", 6, 2) = '08' THEN 'August'
        WHEN substr("Date", 6, 2) = '09' THEN 'September'
        WHEN substr("Date", 6, 2) = '10' THEN 'October'
        WHEN substr("Date", 6, 2) = '11' THEN 'November'
        WHEN substr("Date", 6, 2) = '12' THEN 'December'
        ELSE 'Unknown'
    END AS "Month_Name",
    "Mission_Outcome",
    "Booster_Version",
    "Launch_Site"
FROM
    SPACEXTABLE
WHERE
    substr("Date", 0, 5) = '2015';
```

```
 * sqlite:///my_data1.db
Done.
```

Out[18]:

Month_Name	Mission_Outcome	Booster_Version	Launch_Site
January	Success	F9 v1.1 B1012	CCAFS LC-40
February	Success	F9 v1.1 B1013	CCAFS LC-40
March	Success	F9 v1.1 B1014	CCAFS LC-40
April	Success	F9 v1.1 B1015	CCAFS LC-40
April	Success	F9 v1.1 B1016	CCAFS LC-40
June	Failure (in flight)	F9 v1.1 B1018	CCAFS LC-40
December	Success	F9 FT B1019	CCAFS LC-40

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

Task 10

Rank the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order.

In [19]:

```
%%sql

SELECT
    "Landing_Outcome",
    COUNT(*) AS "Count"
FROM
    SPACEXTABLE
WHERE
    "Date" BETWEEN '2010-06-04' AND '2017-03-20'
GROUP BY
    "Landing_Outcome"
ORDER BY
    COUNT(*) DESC;
```

```
 * sqlite:///my_data1.db
Done.
```

Out[19]:

Landing_Outcome	Count
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

Section 3

Launch Sites Proximities Analysis

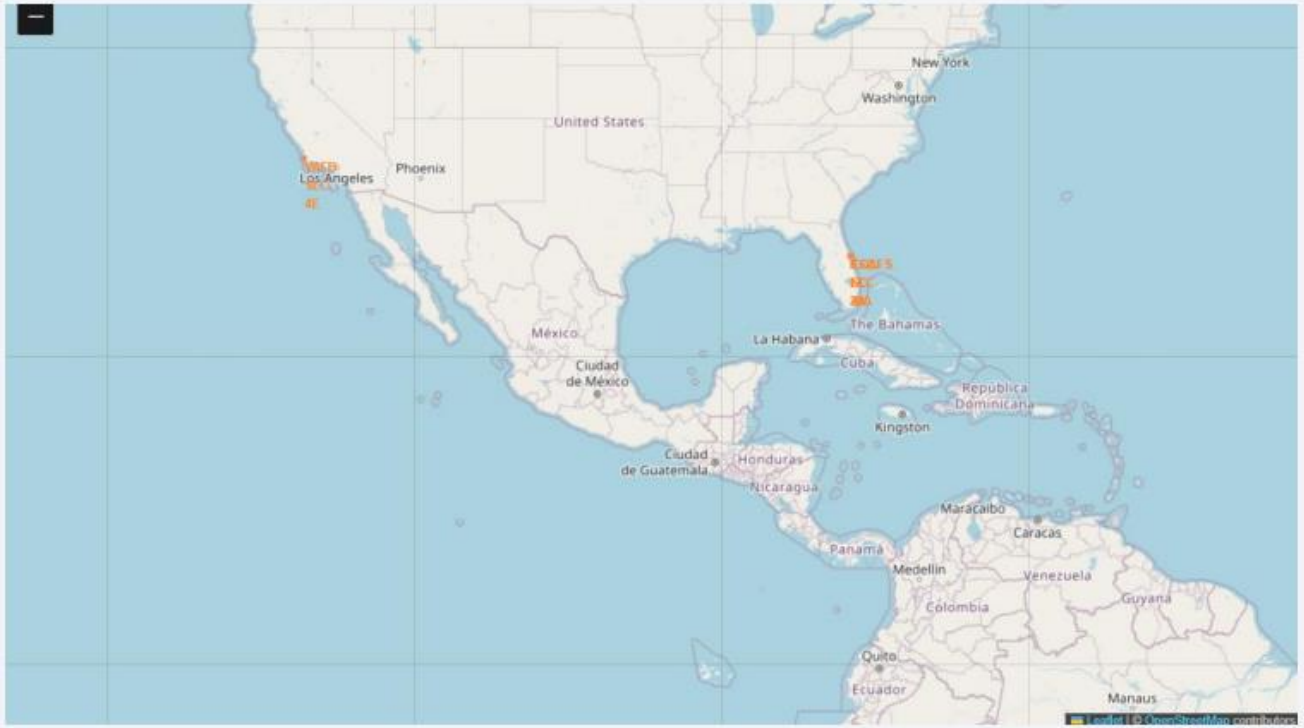
Task 1 Mark all launch sites on a map

1. Are all launch sites in proximity to the Equator line?

- No, not all launch sites are in close proximity to the Equator.
- The launch site at Vandenberg Air Force Base (VAFB SLC4E) is located at a latitude of 34.63, which is further from the Equator compared to the other sites in Florida.

2. Are all launch sites in very close proximity to the coast?

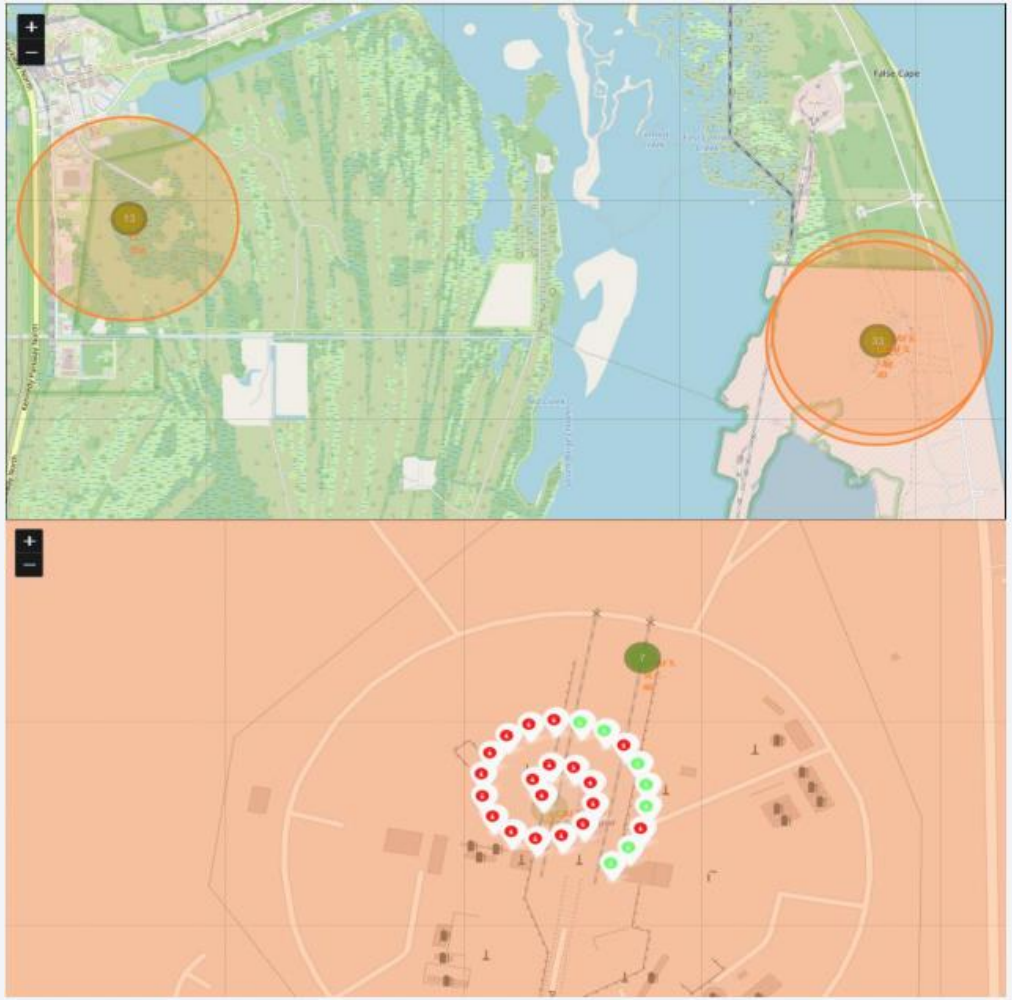
- Yes, all launch sites are in close proximity to the coast.
- The Cape Canaveral sites (CCAFS LC-40 and CCAFS SLC-40) and Kennedy Space Center (KSC LC-39A) are near the coast in Florida.
- Vandenberg Air Force Base (VAFB SLC-4E) is also near the coast in California.



Task 2: Mark the success/failed launches for each site on the map

- This enhanced visualization with clustered markers allows for better exploration and analysis of SpaceX launch data. The clustering makes it easier to manage a large number of markers and observe patterns that might be hidden in a less organized plot. By examining the marker colors and popup information, you can gain deeper insights into the characteristics and distribution of SpaceX launches.

- For example, in the provided screenshot, out of 26 launch sites for CCAFS LC-40, there are 19 red markers and 7 green markers. This color-coding helps to quickly identify the success rate and other categorical distinctions of the launches from this specific site. The red markers might represent unsuccessful launches, while the green markers indicate successful ones, providing immediate visual feedback on the performance of launches at each site.



Task 3: Calculate the distances between a launch site to its proximities

This plot provides a visual representation of the distance between the CCAFS SLC-40 launch site and the closest coastline. The calculated distance is approximately 0.51 kilometers, as indicated by the marker. The added PolyLine clearly shows the straight-line distance, highlighting the proximity of the launch site to the coast. This close proximity to the coastline is typical for launch sites to facilitate over-water flight paths and safe recovery operations, ensuring minimal risk to populated areas.

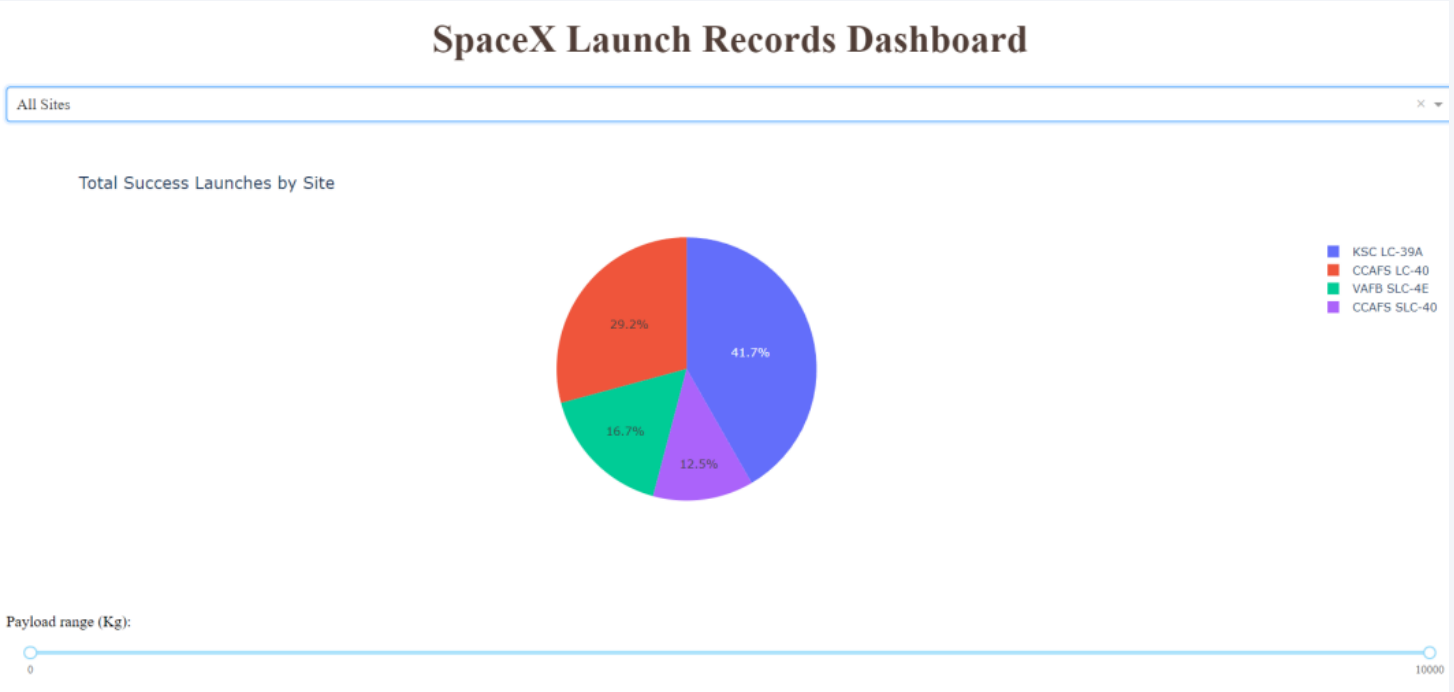
```
coastline_lat = 28.56367
coastline_lon = -80.57163

# Example launch site coordinates (replace with actual launch site coordinates)
launch_site_lat = launch_sites_df.loc[launch_sites_df['Launch Site'] == 'CCAFS SLC-40', 'Lat'].values[0]
launch_site_lon = launch_sites_df.loc[launch_sites_df['Launch Site'] == 'CCAFS SLC-40', 'Long'].values[0]

# Calculate distance using the calculate_distance function
distance_coastline = calculate_distance(launch_site_lat, launch_site_lon, coastline_lat, coastline_lon)

print(f"Distance from launch site to closest coastline: {distance_coastline} km")
```

```
Distance from launch site to closest coastline: 0.5097439631188213 km
```

Section 4

Build a Dashboard with Plotly Dash

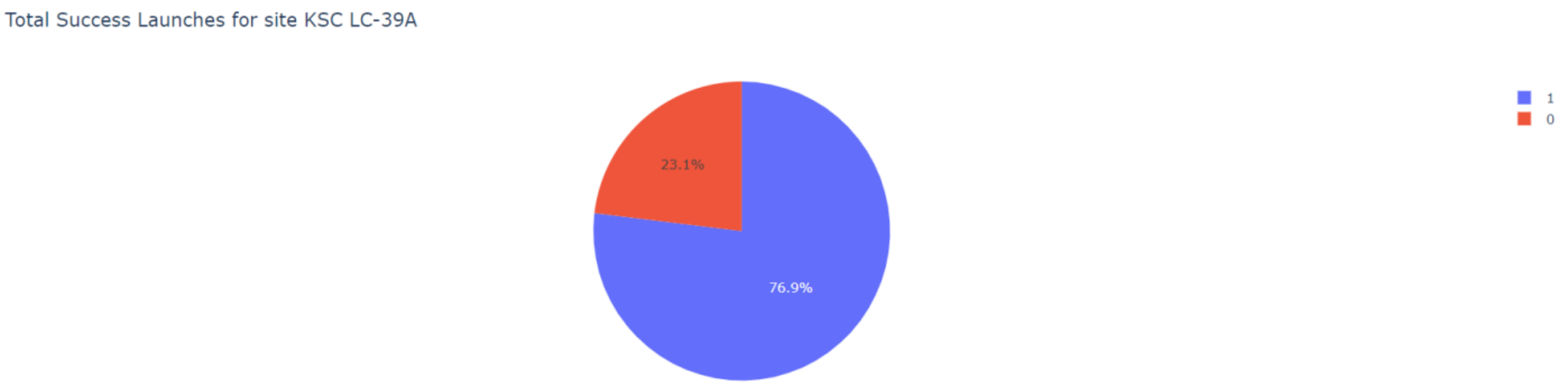
Launch Success Count for all sites (in a pie chart)



Key Findings:

- **CCAFS LC-40**: 29.2%
- **CCAFS SLC-40**: 12.5%
- **VAFB SLC-4E**: 16.7%
- **KSC LC-39A**: 41.7%
- The **KSC LC-39A** launch site has the highest number of successful launches, making up 41.7% of the total successes. This indicates that **KSC LC-39A** is a highly reliable site for SpaceX launches.

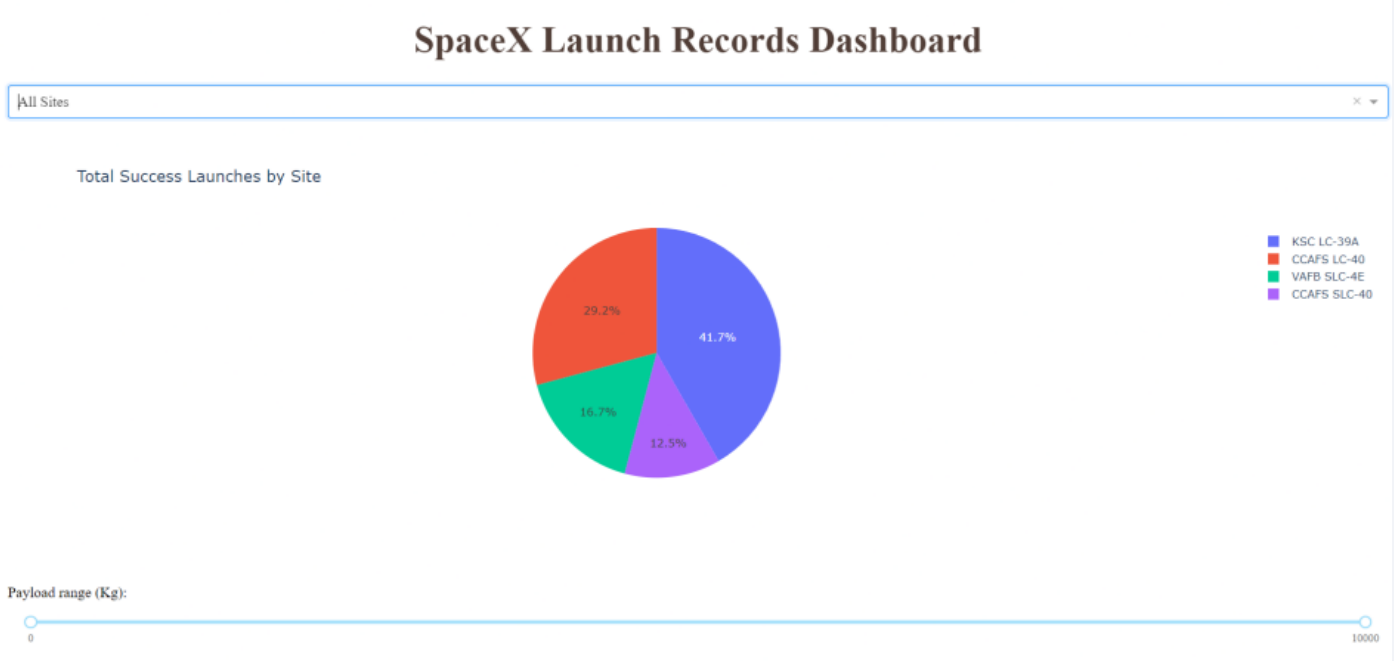
Pie chart for the launch site with highest launch success ratio

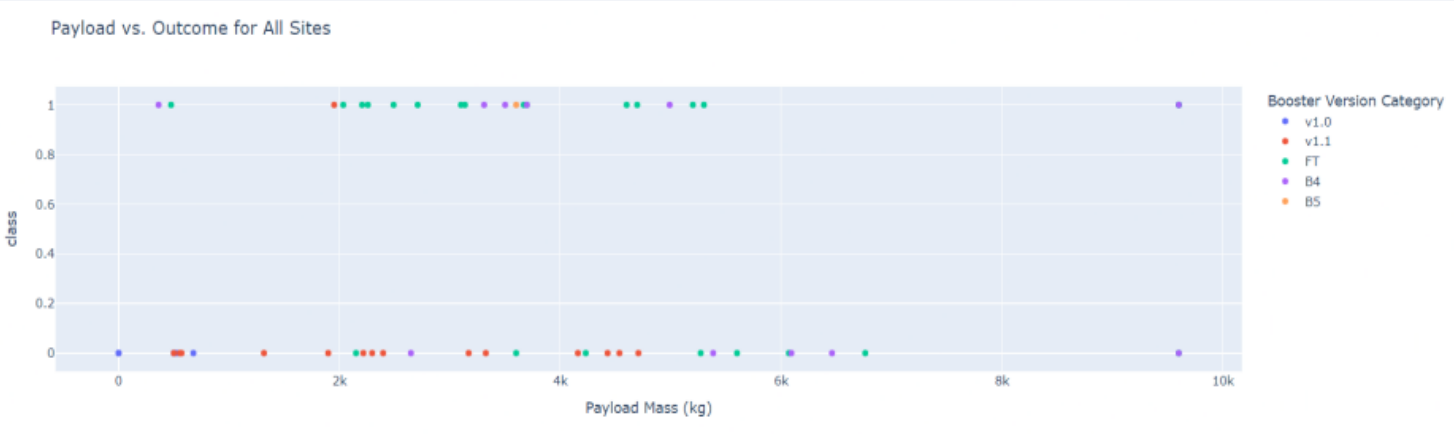


Key Findings:

- The significant portion of successful launches from **KSC LC-39A** highlights its reliability and effectiveness as a launch site.
- For **KSC LC-39A**:
  - **Class 1** (Successful Launches): 76.9%
  - **Class 0** (Unsuccessful Launches): 23.1%
- The high success rate (76.9%) for **Class 1** launches underscores the effectiveness and reliability of the **KSC LC-39A** site.

Key Insights from SpaceX Launch Data Dashboard





Launch Site Success Rates:

- **CCAFS LC-40** has the highest success rate with 43.7% of successful launches.
- This suggests that **CCAFS LC-40** is the most reliable launch site among the ones analyzed.
- Other sites like **KSCLC 39A**, **VAFBSLC-4E**, and **CCAFSSLC-40** have lower success rates, indicating variability in launch success across different sites

Booster Version Performance:

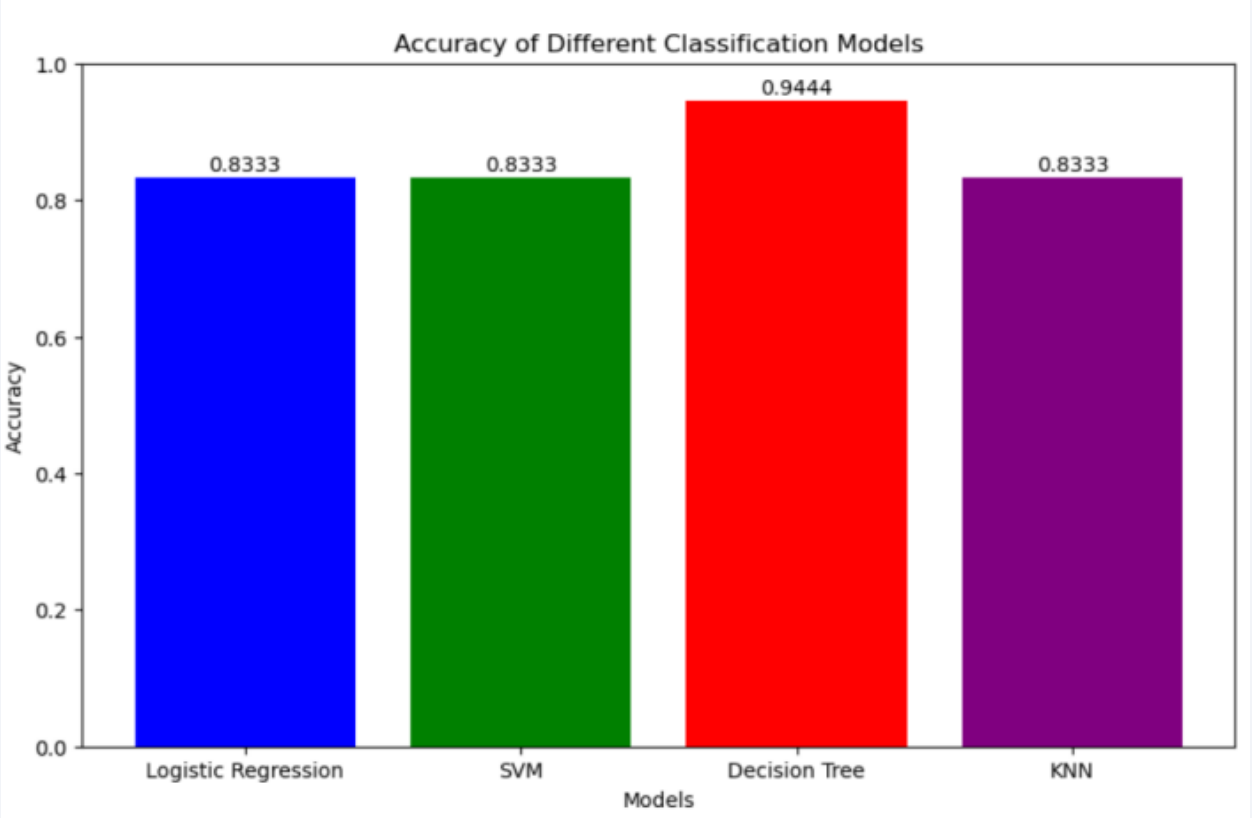
- **Booster version "FT"** appears to be the most frequently used and has a high success rate across various payload masses.
- **Booster version "v1.0"** has fewer launches and may require further analysis to understand its performance.
- Overall, booster versions do not show a clear trend that higher payload masses correlate with lower success rates.

Section 5

Predictive Analysis (Classification)

Classification Accuracy

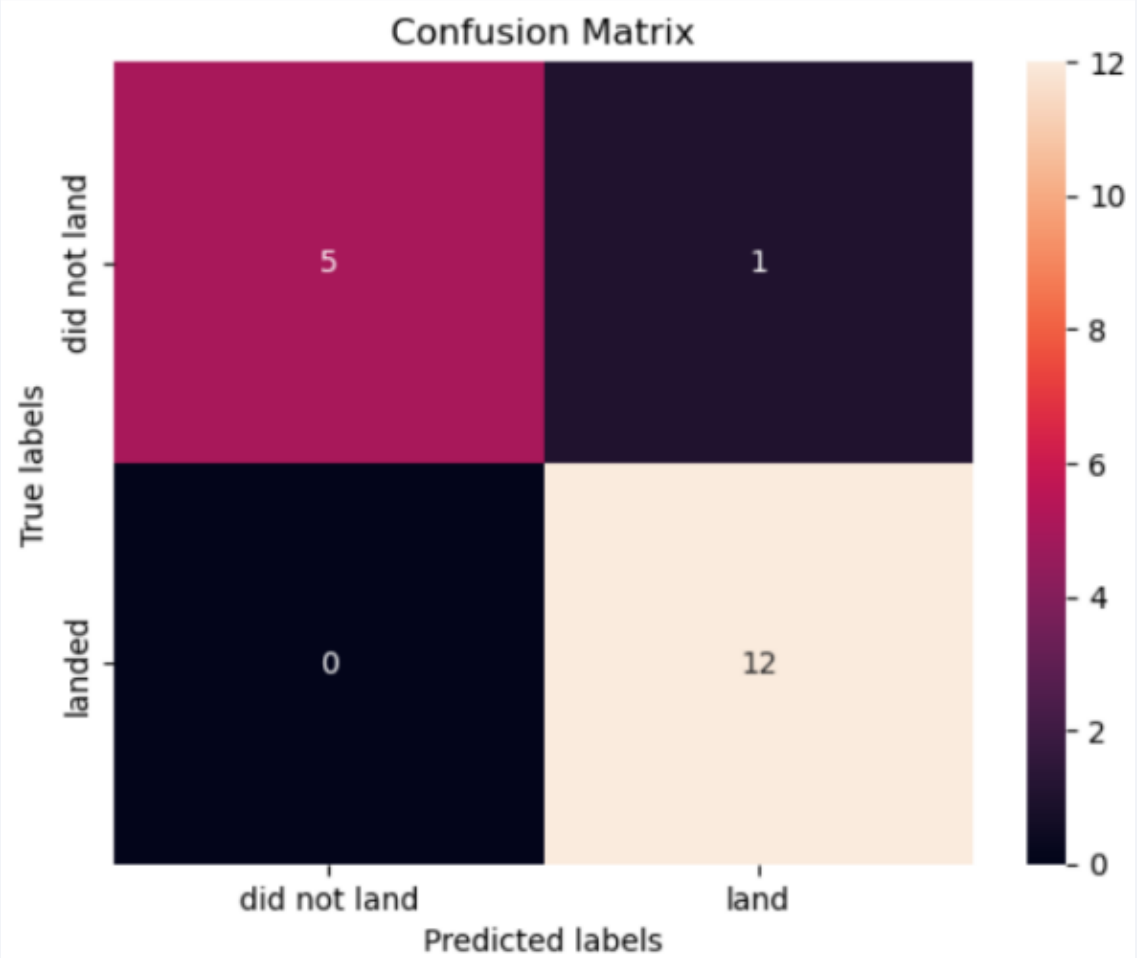
- Based on the results, the Decision Tree model has the highest classification accuracy on the test data, achieving an accuracy of 0.9444. This suggests that the Decision Tree model is better suited for this dataset compared to Logistic Regression, Support Vector Machine, and K Nearest Neighbors, all of which achieved an accuracy of 0.8333.



Confusion Matrix

Explanation and Insights

- **High Accuracy**: The model achieved a high accuracy score of 94.44%, with a significant number of true positives and true negatives, demonstrating its effectiveness in predicting Falcon 9 first stage landings.
- **No False Negatives**: The absence of false negatives indicates that the model reliably predicts successful landings. This is crucial for ensuring readiness and safety in aerospace operations, as every actual successful landing was accurately identified.
- **Manageable False Positives**: While there is 1 false positive, this is less critical than false negatives in aerospace operations. Over -preparation (due to false positives) is more manageable than under -preparation, making the model's performance highly acceptable for practical applications.
- **Balanced Performance**: The model shows a balanced performance with a slight bias towards predicting successful landings. This aligns well with practical needs in the aerospace industry, where ensuring successful landings is of paramount importance for cost estimation and planning.



Conclusions

- **Point 1:** Data evaluation highlights "CCAFS LC-40" as the leading launch facility, securing a 43.7% share of all successful missions. This suggests that the site's operational protocols or environmental factors are exceptionally well-suited for achieving positive outcomes.

- **Point 2:** Evidence from scatter plot visualizations indicates that the "FT" booster variant maintains a consistently high success frequency across a broad spectrum of payload weights. Its proven reliability makes it a superior choice for ensuring mission integrity compared to earlier iterations.

- **Point 3:** The study found no statistically significant correlation between increased payload mass and mission failure. Consequently, success appears more dependent on variables such as specific booster performance and launch site logistics rather than the weight of the cargo itself.

Conclusions

- **Point 4:** Interactive spatial mapping and real-time data visualizations provided a comprehensive overview of SpaceX's launch landscape. By leveraging Folium and Dash, we bridged the gap between raw data and strategic decision-making, allowing for more precise operational assessments.

- **Conclusion:** Ultimately, this dual approach of advanced analytics and interactive reporting has not only identified the levers of success for SpaceX but also created a reliable framework for industry benchmarking. The resulting insights are instrumental for optimizing launch strategies and securing the long-term viability of cost-effective rocket reuse.

Appendix

- IBM: For providing the course and learning materials.
- Coursera: For the platform to access and complete the course.

Thank you!